FENOC™
FirstEnergy Nuclear Operating Company

Perry Nuclear Power Plant
P.O. Box 97
10 Center Road
Perry, Ohio 44081

David B. Hamilton
Vice President

440-280-5382

June 26, 2017
L-17-204

10CFR50.73(a)(2)(v)

ATTN: Document Control Desk
U. S. Nuclear Regulatory Commission
Washington, DC 20555-0001

SUBJECT:
Perry Nuclear Power Plant
Docket No. 50-440, License No. NPF-58
Licensee Event Report Submittal

Enclosed is Licensee Event Report (LER) 2017-003, "Annulus Exhaust Gas Treatment System Loss of Safety Function".

If there are any questions or if additional information is required, please contact Mr. Nicola F. Conicella, Manager – Regulatory Compliance, at 440-280-5415.

Sincerely,

David B. Hamilton

Enclosure:
LER 2017-003

cc: NRC Project Manager
NRC Resident Inspector
NRC Region III

NRC FORM 366 (04-2017) U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/

APPROVED BY OMB: NO. 3150-0104 EXPIRES: 3/31/2020

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. PAGE
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4. TITLE:
Annulus Exhaust Gas Treatment System Loss of Safety Function

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER 05000
04	27	2017	2017 -	003 -	00	06	26	2017	FACILITY NAME	DOCKET NUMBER 05000

9. OPERATING MODE: 1

10. POWER LEVEL: 100

11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: *(Check all that apply)*

☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☒ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ OTHER Specify in Abstract below or in NRC Form 366A	

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT:	TELEPHONE NUMBER *(Include Area Code)*
David Lockwood – Regulatory Compliance	(440) 280-5200

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX
A	BH	DMP	X999	Y					

14. SUPPLEMENTAL REPORT EXPECTED	15. EXPECTED SUBMISSION DATE	MONTH	DAY	YEAR
☐ YES *(If yes, complete 15. EXPECTED SUBMISSION DATE)* ☒ NO				

ABSTRACT *(Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)*

On April 27, 2017, at 0545 hours, with the plant in Mode 1 at 100 percent rated thermal power, while shifting Annulus Exhaust Gas Treatment System (AEGTS) from sub-system B to sub-system A, annulus differential pressure could not be maintained within the required system operating band, which caused an unplanned entry into technical specification limiting conditions of operation and a momentary loss of safety function. Sub-system B was inoperable while being shutdown to standby readiness in accordance with plant operating procedures and would not have automatically started in response to an actuation signal. Investigation determined that sub-system A was inoperable due to failed recirculation damper. Sub-system B was restarted at 0550 hours, to maintain annulus differential pressure. Annulus differential pressure was maintained above the minimum differential pressure throughout the event and the technical specification limiting conditions for operation were not exceeded.

The cause for the AEGTS sub-system A recirculation damper failure was a failure to follow procedure which resulted in the split coupling that connects the actuator to the damper not being properly tightened. The recirculation damper was repaired and AEGTS sub-system A was restored to operable status on April 28, 2017, at 2208 hours. The failure to follow procedure was addressed by the FENOC performance management process.

Since AEGTS is not a core damage mitigation system and does not mitigate large and early containment releases, the inoperability of the AEGTS is determined to be of small safety significance. This event is being reported in accordance with 10CFR50.73(a)(2)(v)(C) and 10CFR50.73(a)(2)(v)(D) as an event or condition that could have prevented the fulfilment of a safety function.

APPROVED BY OMB: NO. 3150-0104 EXPIRES: 3/31/2020

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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Perry Nuclear Power Plant	05000- 440	2017	- 003	- 00

NARRATIVE

Energy Industry Identification System (EIIS) codes are identified in the text as [XX].

INTRODUCTION

On April 27, 2017, at 0545 hours, with the plant in Mode 1 at 100 percent rated thermal power, while shifting Annulus Exhaust Gas Treatment System (AEGTS) [BH] from sub-system B to sub-system A, annulus differential pressure could not be maintained within the required system operating band, which caused an unplanned entry into technical specification limiting conditions of operation and a momentary loss of safety function. Sub-system B was inoperable while being shutdown to standby readiness in accordance with plant operating procedures and would not have automatically started in response to an actuation signal. Investigation determined that sub-system A was inoperable due to a failed recirculation damper [DMP]. Sub-system B was restarted at 0550 hours, to maintain annulus differential pressure. Annulus differential pressure was maintained above the minimum differential pressure throughout the event and technical specification limiting conditions for operation were not exceeded.

The recirculation damper was repaired and AEGTS sub-system A was restored to operable status on April 28, 2017, at 2208 hours.

DESCRIPTION OF EVENT

AEGTS consists of two 100 percent sub-systems. AEGTS processes the ambient air in the annular space between the shield building and the containment vessel to limit a release to the environment of radioisotopes which may leak from the primary containment under accident conditions. The AEGTS is a recirculation type system with split flow. Some of the filtered air extracted from the annulus space is recirculated and some is discharged to the unit vent. AEGTS is designed to continuously maintain a negative pressure differential of 0.25 inches of water gauge minimum between the containment vessel annulus and the outside environment and direct the exhaust flow from the annulus through a charcoal filter plenum to ensure that the release of radioactivity to the environment is below permissible discharge limits. During normal operation the expected discharge to the unit vent is approximately 700 cfm, at a negative pressure differential of 0.66 inches of water gauge. The 0.66 inches of water gauge pressure differential is provided to maintain the 0.25 inches of water gauge minimum pressure differential required due to instrument location, to meet plant post-LOCA conditions, and to adjust for all environmental conditions. Operating parameters in accordance with plant procedures require annulus pressure to be maintained between 0.75 inches and 1.0 inches water gauge negative pressure differential.

AEGTS sub-system A was inoperable from April 24, 2017, at 0427 hours, until April 26, 2017, at 1453 hours, for planned maintenance which included replacement of the actuator on the recirculation damper.

On April 27, 2017, while shifting AEGTS from sub-system B to sub-system A, annulus differential pressure could not be maintained within the required system operational band. During the shifting operation the off going AEGTS sub-system (in this event, B) exhaust fan's switch is held in STOP prior to being placed in STANDBY, which interrupts current flow to a seal-in circuit and stops the exhaust fan. If an actuation signal is received while holding exhaust fan switch in the STOP position, the actuation signal is overridden, and another seal-in circuit will continue to override the actuation signal when the switch is placed in STANDBY. Therefore, the off going AEGTS subsystem is declared inoperable while holding the exhaust fan switch in STOP, in accordance with plant operating procedures. AEGTS sub-system B was declared inoperable at 0544 hours while the exhaust fan switch was held in STOP and declared operable at 0545 hours following release of the switch.

Control room operators recognized that annulus differential pressure was outside of the operating band (0.7 inches water gauge negative differential pressure) and not improving. Control room operators started AEGTS sub-system B at 0550 hours, and stopped AEGTS sub-system A at 0556 hours. AEGTS sub-system A was declared inoperable at 0545 hours. Investigation determined that sub-system A was inoperable due to a failed recirculation damper.

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NARRATIVE

CAUSE OF EVENT

The cause for the AEGTS sub-system A recirculation damper failure was a failure to follow procedure by electrical, and instrument and controls technicians which resulted in the split coupling that connects the actuator to the damper not being properly tightened.

ANALYSIS OF EVENT

Since AEGTS is not a core damage mitigation system and does not mitigate large and early containment releases, the inoperability of the AEGTS is determined to be of small safety significance. This event is being reported in accordance with 10CFR50.73(a)(2)(v)(C) and 10CFR50.73(a)(2)(v)(D) as an event or condition that could have prevented the fulfilment of a safety function.

CORRECTIVE ACTIONS

The failure to follow procedure was addressed by the FENOC performance management process.

PREVIOUS SIMILAR EVENTS

A review of LERs and the corrective action database for the past three years identified that no previous similar events occurred involving AEGTS.

COMMITMENTS

There are no regulatory commitments contained in this report. Actions described in this document represent intended or planned actions, are described for the NRC's information, and are not regulatory commitments.